

EXPERIMENT 13

Generate a parse tree for a given sentence using a context-free grammar using python program.

PROGRAM:

```
import nltk

from nltk import CFG

from nltk.parse import RecursiveDescentParser

grammar = CFG.fromstring("""

S -> NP VP

NP -> Det N

VP -> V NP

Det -> 'the'

N -> 'boy' | 'apple'

V -> 'eats'

""")

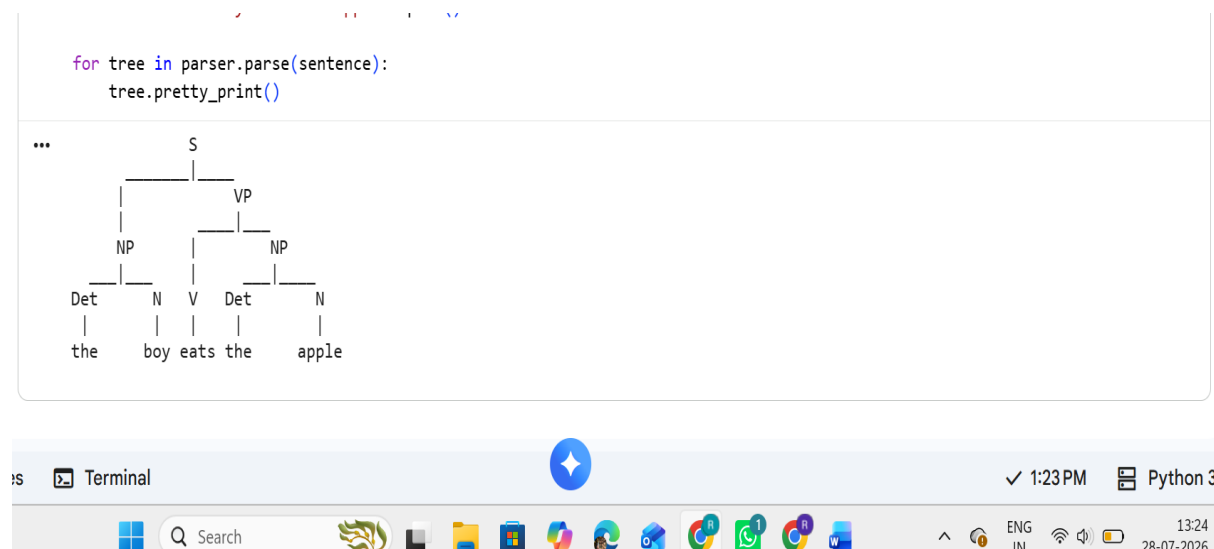
parser = RecursiveDescentParser(grammar)

sentence = "the boy eats the apple".split()

for tree in parser.parse(sentence):

    tree.pretty_print()
```

OUTPUT:



The screenshot shows a terminal window with the following output:

```
...
      S
     / \
    NP  VP
   / \ / \
  Det N V Det N
  |  | | | |
the boy eats the apple
```

The terminal window also displays the code used to generate the parse tree, which is the same as the program shown in the previous block. The window title is "Terminal" and the system tray shows the time as 1:23 PM and the date as 28-07-2026.